Progeam - D

16 bit multiplication & division

given

multiplication : 1357H x 0025H

Division : 9A23H ÷ 0011H

Multiply

1357H x 0025H = 0002 CA13H

Upper word = 0002H

Lower word = CA13H

Division :

9A23H ÷ 00

Quo = 0911

Rem = 0002

EXP - 2

Progeam: A

AND, OR, XOR and Nibble making

given X = ~~47~~ AH, Y = 3CH, W = 1234H

(i) AND operation

X = AFH = 1010 0111

Y = 3CH = 0011 1100

0010 0100

X AND Y = 24H

(ii) ORH operation.

X = AFH = 1010 0111

Y = 3CH = 0011 1100

1011 1111

X OR Y = BFH

3) XOR operation:-

X = A7H = 1010 0111

Y = 3CH = 0011 1100

1001 1011

X XOR Y = 9BH

4) Lower Nibble Masking :-

Mask = 0FH = 0000 1111

X = 1010 0111

Mask 0000 1111

Result = 0000 0111

LNM = 07H

5) Upper Nibble Masking

Mask = F0H = 1111 0000

X = 1010 0111

Mask 1111 0000

Result = 1010 0000

Result = A0H

6) One's complement of X

X = A7H = 1010 0111

$\therefore X' = 0101\ 1000$

Result = 58H

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7) 2' complement of X

Two's complement = One's complement + 1

= 58 H

+ 01 H

59 H

Verification:-

A7H

59 H

100 H

8) one's complement of 16 bit W

W = 1234 H

∴ W = 0001 0010 0011 0100

W' = 1110 1101 1100 1011

∴ Result: EDCB H

9) Two's complement of 16 bit W

∴ one's complement = EDCB H

+0001 H

EDCCH

Verification:-

1234 H + EDCCH = 10000H

10) Flag calculation After AND

Result = 24H = 0010 0100

S = 0, Z = 0, AC = 1, P = 1, CY = 0

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Program - B

Given D = CAH = 1100 1010

W = 9234H

(1) Set Bits 7 and 0

Mask = 81 H = 1000 0001

D = 1100 1010

Mask = 1000 0001

D OR 81H = 1100 1011

Result = CBH

2) Clear Bits 3 and 1

Mask = F5H = 1111 0101

D = 1100 1010

Mask = 1111 0101

D AND F5H = 1100 0000

Result = C0 H

3) Toggle lower Nibble

Mask = 0FH = 0000 1111

D = 1100 1010

Mask = 0000 1111

D XOR 0FH = 1100 0101

Result = C5H

4) Test Bit 6

Mask = 40H = 0100 0000

D = 1100 1010

Mask = 0100 0000

0100 0000

= 40H

∴ Result is stored as 01H

5) Rotate left (RLC)

D = 1100 1010

Ori = 1100 1010

Rot = 1001 0101

∴ Result = 95H

CY = 1

6) Rotate Right (RRC)

Ori = 1100 1010

Rot = 0110 0101

∴ Result = 65H

CY = 0

7) Logical shift left.

ori = 1100 1010

shifted = 1001 0100

∴ Result = 94H

CY = 1

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8) Logical Shift Right

Ori = 1100 1010

Shifted = 0110 0101

∴ Result = 65 H

CY = 0

9) Arithmetic Shift Right

ori = 1100 1010

Shifted = 1110 0101

Result = E5H , CY = 0.

signed interpretation = −54 ÷2

= −27

10) Count the no. of 1's

D = 1100 1010

∴ Number of 1s = 4

11) 16 bit shift left

given W = 9234 H

9234H = 1001 0010 0011 0100

shift = 0010 0100 0110 1000

Result = 2468 H

CY = 1

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